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Date: 1/15/24

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Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find the gradient of the function $f(x,y) = 3y - 4x$ at the point $(3, 5)$. Then sketch the gradient and the level curve that passes through that point.

The x- and y-components of ∇f are $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, respectively. To find the gradient, start by finding both partial derivatives of the function.

Each partial is shown below.

$$\frac{\partial f}{\partial x} = -4 \quad \frac{\partial f}{\partial y} = 3$$

Use $\frac{\partial f}{\partial x} = -4$ and $\frac{\partial f}{\partial y} = 3$ to find ∇f . The values of the partial derivatives correspond to the components of ∇f .

$$\nabla f = -4\mathbf{i} + 3\mathbf{j}$$

The level curve is the set of points in a plane where a function $f(x,y)$ has a constant value, $f(x,y) = c$. To find the level curve, find the value of $f(x,y)$ at the point $(3, 5)$ by substituting the x- and y-coordinates of that point into $f(x,y)$, $3y - 4x$.

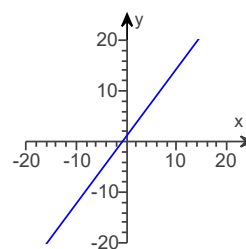
$$\begin{aligned} f(3,5) &= (3)(5) - (4)(3) \\ f(3,5) &= 3 \end{aligned}$$

To sketch the level curve, rewrite the given equation so that y is expressed as a function of x, where $f(x,y)$ is equal to a constant, in this case 3. Solve the resulting equation for y.

$$\begin{aligned} y &= \frac{4x + f(x,y)}{3} \\ y &= \frac{4x + 3}{3} \end{aligned}$$

Note that the expression $y = \frac{4x + 3}{3}$ is in the form $y = mx + b$. Therefore, the level curve at the point $(3, 5)$

is a straight line with a slope of $\frac{4}{3}$ and a y-intercept of 1.



The gradient vector will always be perpendicular to the level curve and its base will always be at a point on the curve, in this case, $(3, 5)$. To find its head, add the components of the gradient to the coordinates of the point. In this case, the head of the gradient $-4\mathbf{i} + 3\mathbf{j}$ will be at the point $(-1, 8)$.

